



Olanipekun Adekunle

Senior Backend Engineer
(Platform & Backend Systems)

CONTACT INFO:

+234-7039365725
adekunle.olanipekun.ko@gmail.com
Lagos , Nigeria.
www.github.com/kunle001
https://www.linkedin.com/in/kunle-olanipekun-555764225/

SUMMARY

Backend & Infrastructure Engineer with 6+ years of experience designing and scaling distributed systems in fintech and banking. Expert in Golang, Node.js, and cloud-native infrastructure (Kubernetes, CI/CD), with a strong focus on reliability, performance, and maintainability. Led key initiatives including large-scale data migrations, payment systems, and identity platforms, driving architectural decisions and influencing system design across teams. Known for solving complex system challenges, improving reliability, and mentoring engineers to uphold high engineering standards.

SKILLS

Languages & Frameworks

- Node.js, TypeScript, JavaScript, Golang.
- Backend & Architecture
- Microservices, Distributed Systems, REST APIs, gRPC, WebSockets,
- Clean Architecture (Hexagonal, Domain Driven designs)

Databases & Storage

- PostgreSQL, MongoDB, MySQL, MSSQL, Database Migration & Optimization

DevOps & Infrastructure

- Kubernetes (Production), Docker, CI/CD (GitLab/GitHub Actions),
- ArgoCD (GitOps), Infrastructure as Code, Self-Managed Cloud,
- Monitoring (Prometheus, Grafana)

Cloud Platforms

- AWS, Azure, On-Premise Infrastructure (Linux and Windows)

Fintech & Domain Expertise

- Payment Systems, Core Banking, KYC/Identity Verification,
- Transaction Processing, Financial Compliance (CBN),
- Bill Payment Integration
- Real-Time & Performance
- WebRTC, WebSockets, High-Throughput Systems,
- Concurrency Patterns, Performance Optimization

Leadership & Practices

- Technical Leadership, Code Review & Mentoring,

ACADEMIC BACKGROUND

BACHELOR OF ENGINEERING

FEDERAL UNIVERSITY OF AGRICULTURE ABEOKUTA | 2023

NATIONAL SOCIETY OF ENGINEERS (NSE MEMBER).

NATIONAL YOUTH SERVICE CORPS (NYSC).

EXPERIENCE

SLOCX.

Team Lead.

Aberdeen, Scotland, United Kingdom
2026 - Present

- Led the design and roll-out of an in-lesson class-tool (realtime whiteboard, tutor content library, vocabulary practice, live quizzes) turning Sloxc from "video call + booking" into a full teaching workspace, increasing tutor session retention by 80% and reducing per-lesson drop-off by 80%.
- Integrated WayForPay end-to-end with idempotent settlement and HMAC-verified webhooks, cutting failed payment reconciliations by 45% and eliminating 100+ monthly manual finance postings.
- Executed an AWS s3 bucket → self-hosted MinIO migration of user-generated media via a resumable zero-downtime migrator, reducing third-party media spend by 10% MoM while retaining full data locality.
- Re-platformed the ecosystem onto self-hosted Kubernetes with Helm + ArgoCD GitOps, cutting deployment time from 10 to under 2 minutes and enabling deterministic cluster stand-up from a single sync.
- Scaled the tutor lifecycle system vetting, scheduling, attendance-driven splits, and payouts to onboard 600+ tutors and 1,000+ students across 6 languages, unlocking multi-region expansion.
- Established a supply-chain security program (full-history secret scanning, image vulnerability gates, credential rotation across 5 providers), reducing exposed-secret incidents to zero and blocking HIGH/CRITICAL vulnerabilities from reaching production.
- Rebuilt the admin operations surface paginated filtered users, CSV exports, tutor payout workflow, off-channel payment reconciliation reducing admin ticket resolution time by 60% and enabling finance ops without engineering intervention.
- Led a 9-engineer team through the transition from single-VM hosting to a GitOps-managed Kubernetes fleet, codifying hiring bar, dependency-addition policy, and cross-service design-review practice.

STERLING BANK (HQ)

Senior Backend Engineer.

Marina Lagos, Nigeria
2024- 2026

- Led backend development of KYC, identity verification, and payment workflows for regulated financial products, strengthening fraud prevention and ensuring full regulatory compliance.
- Designed and implemented a robust encryption layer for all API communications, enabling secure request/response handling and safeguarding financial transactions.
- Redesigned the card transaction service to enforce strict pre-transaction validations, eliminating critical loopholes and preventing significant financial losses.
- Spearheaded the migration of customers from Bankly (acquired by Sterling) to Gomoney, delivering a seamless transition and enabling users to effortlessly reclaim their accounts.
- Resolved legacy system issues and optimized performance, significantly improving application stability and reliability.
- Built internal admin, operations, and compliance tools that automated key workflows, reducing customer issue volume by approximately 80%.
- Integrated multiple external fintech vendors with failover and resiliency strategies, improving system availability and operational continuity.
- Implemented CBN-compliant address verification (OkHi), reducing onboarding risk and enhancing regulatory adherence.
- Collaborated closely with Compliance, Risk, Operations, and Support teams to resolve high-severity production issues and eliminate recurring defects.
- Led migration to a new KYC provider, significantly improving onboarding success rates and reducing related issues by ~80%.

NEPTUNE MICROFINANCE BANK.

Senior Backend/ Lead Infrastructure Engineer

Ikeja Lagos, Nigeria
2022 - 2025

- Contributed to architecture and delivery of a microservices Core Banking Platform, improving scalability and reducing release cycles across key banking domains.
- Led infrastructure and CI/CD setup for POS deployments, strengthening reliability and speeding up production releases.
- Setup and operated self-managed data and messaging infrastructure (Kafka, RabbitMQ, Redis, MongoDB, MySQL, PostgreSQL), improving resiliency and reducing vendor dependency.
- Migrated workloads to an on-prem Kubernetes environment, saving \$50K+ annually while maintaining performance SLAs.
- Led backend development of a Corporate Web Banking Platform with KYC, RBAC, and transaction workflows, driving enterprise digital adoption.
- Designed a secure API Gateway (gRPC/REST) that accelerated partner integrations and standardized service access.
- Delivered a multi-channel notification service (1K+ alerts/day) improving customer engagement and operational monitoring.
- Directed legacy customer data migration with near-zero downtime, ensuring data integrity during platform transition.
- Implemented identity verification and payments integrations (BVN, NIN, Interswitch, Providus) with failover, improving success rates.
- Built a secure 2FA e-Token platform enhancing fraud prevention and compliance.
- Mentored engineers and enforced best practices, improving code quality and team velocity.

KHAIME AI

Team Lead (Backend)

Texas, USA

2021 - 2022

- Took ownership of a poorly architected codebase, redesigning data models, refactoring database schemas, and introducing a clean, modular architecture optimized for scalability and long-term maintainability.
- Designed and executed large-scale data transformation and migration pipelines with full backup strategies, ensuring zero data loss and strong consistency across newly structured schemas.
- Built a robust recurring payment scheduling system, enabling automated payments for goods and services with support for edge cases, retry mechanisms, and idempotent execution.
- Decomposed a monolithic backend into independently deployable microservices, enabling per-service scalability, improved fault isolation, and faster release cycles.
- Optimized high-traffic API endpoints, reducing response times by over 60% through advanced query tuning, caching strategies, and asynchronous processing.
- Led stakeholder planning sessions, translating business requirements into clear technical roadmaps and driving execution to delivery.
- Mentored junior engineers on backend architecture, coding standards, and development workflows, establishing best practices adopted across the team.

COLLAB CARDS.

Backend Engineer (BACKEND)

United Kingdom

2019- 2020

- Architected and collaborated in the development of a scalable Node.js backend for a social content platform (writers, poets, fiction), delivering core features including content publishing, likes, comments, and idea sharing on PostgreSQL, Docker, and AWS.
- Designed high-performance REST APIs using Express.js and clean architecture principles, supporting user management, bookings, and real-time chat services with consistent sub-100ms response times.
- Engineered a low-latency real-time messaging system using Socket.io, leveraging Node.js event-driven, non-blocking I/O and worker threads to efficiently handle high volumes of concurrent connections.
- Optimized PostgreSQL performance and implemented Redis caching strategies, significantly reducing latency and improving throughput under high concurrent, multi-region traffic.
- Integrated multiple third-party services (payments, analytics, email) and implemented CI/CD pipelines with GitLab and Docker, automating testing and cloud deployments for faster, reliable releases.
- Established centralized observability using structured logging (Winston/Pino), metrics, and alerting systems, improving system visibility and reducing incident response time.
- Built comprehensive automated test suites with Jest and Supertest, and collaborated cross-functionally with frontend, product, and QA teams to deliver secure, production-ready features.

PROJECTS

GOMONEY

<https://play.google.com/store/apps/details?id=com.global.gomoney>

Contributed to the development of Gomoney, a modern fintech app offering seamless bill payments, peer-to-peer transfers, virtual cards, budgeting tools, transaction categorization, spending analytics, and in-app customer support — designed to enhance user financial management and digital banking experiences.

NVAULT

<https://play.google.com/store/apps/details?id=com.neptune.nvault&hl=en>

Contributed to the development of nVAULT, a cutting-edge fintech app at Neptune Microfinance Bank Ltd., revolutionizing traditional banking with an all-in-one digital platform for payments, savings, and investments. Played a key role in designing a streamlined digital onboarding process, enhancing user experience, and optimizing savings and investment features to drive user engagement and financial empowerment.

QUIZMEAPP

<https://play.google.com/store/apps/details?id=com.quizme.quizme>

Led the development of a rewards-based gaming app where users earn by answering questions, gaining points, and competing on a leaderboard. Features included cash withdrawals, point redemption for airtime and data, and real-time performance tracking.

ZERO PAY

<https://play.google.com/store/apps/details?id=com.myzeropaypay.app>

Led the Development a real estate application with flexible installment payments, secure wallet integration, and automated direct debit mandates. The app also supports utility payments (airtime, data, electricity), offering users a seamless property purchase and financial management experience.

SLOCX

<https://slocx.com>

Lead a Team of 9 to develop online language center for learning languages in over five countries offers various features, such as live chatting, live video lessons with tutors, lesson booking, and more. The backend is built using Golang, a highly performant and scalable language ideal for handling concurrent tasks. The web framework Chi is used to develop lightweight and fast REST APIs for handling various functionalities, such as user management, lesson booking, and chat services.

For real-time communication, such as live video lessons and messaging, WebSockets are utilized on the backend. They facilitate a persistent connection between the server and client, ensuring low-latency communication essential for interactive features.

On the frontend, WebRTC is integrated for seamless, peer-to-peer video streaming, allowing students to engage in live video lessons with their tutors. This combination of technologies ensures a smooth, real-time interactive learning experience across multiple regions.

BEAST MOOD.

beastmood-anime.vercel.app

Led a Team of Backend Engineers in developing an **anime streaming platform** that offers features like real-time video streaming, user comments, and the ability to like content. The platform is built with **Node.js** and **Express** to handle server-side logic, while **MongoDB** provides flexible and scalable data storage for user interactions, such as comments and likes. The project is hosted on **AWS**, ensuring reliable performance and scalability. Although the core functionality is in place, the platform is still in development, with additional features and optimizations being worked on to enhance the user experience.